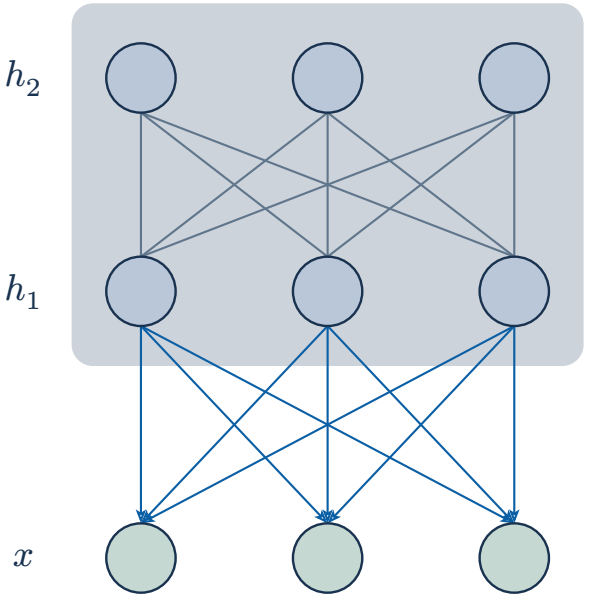

Undirected top pair

RBM prior: $p(h_1, h_2)$

Downward generation

Conditional model: $p(x \mid h_1)$

$$p(x, h_1, h_2) = p(h_1, h_2)p(x \mid h_1)$$